

Department of Computer Science and Engineering
University College of Engineering (A)
Osmania University

B.E V-Semester WP (CSE_AIML) Class Test-II-2026

Subject: **Machine Learning**

Time: 6:00 to 7.30 PM .

Max. Marks: 30

Date: 28-03-2026

Note: Answer all Question in Part-A and any two in Part-B

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Q.NO	Part –A (5 x 2 = 10)										Marks	BT	CO																																	
1	Describe Perceptron and any two activation functions										2	1	1																																	
2	Consider a coin that is known to be either fair or 60% in favour of heads. Determine which kind of coin it is based on the observations, given $P(h_1) = 0.75$ and $P(h_2) = 0.25$										2	1	1																																	
3	Describe the following (i) Regularization (ii) K-Fold cross Validation										2	1	1																																	
4	Briefly describe the steps to calculate a lower dimensional subspace of the LDA technique.										2	1	1																																	
5	What is Reinforcement Learning? What are the elements of Reinforcement Learning?										2	1	1																																	
Part –B (2 x 10 = 20)																																														
6 (a)	What is support vector? Explain how Support Vector Machine works with suitable examples										5	3	2																																	
(b)	What are SVM Kernels. Write notes on (any two) (i) Linear Kernel (ii) Polynomial Kernel (iii) Radial basis function kernel (iv) Sigmoid kernel										5	3	2																																	
7 (a)	A study is conducted involving 10 patients to investigate the relationship and effects of patients age and the blood pressure. Calculate the linear regression of patients age and blood pressure <table border="1"><tr><td></td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>Age</td><td>35</td><td>40</td><td>38</td><td>44</td><td>67</td><td>64</td><td>59</td><td>69</td><td>25</td><td>50</td></tr><tr><td>BP</td><td>112</td><td>128</td><td>130</td><td>138</td><td>158</td><td>162</td><td>140</td><td>175</td><td>125</td><td>142</td></tr></table>											1	2	3	4	5	6	7	8	9	10	Age	35	40	38	44	67	64	59	69	25	50	BP	112	128	130	138	158	162	140	175	125	142	8	6	2
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Age	35	40	38	44	67	64	59	69	25	50																																				
BP	112	128	130	138	158	162	140	175	125	142																																				
(b)	Write short notes on Back propagation										2	1	1																																	
8 (a)	Write notes on (i) K-Means algorithm (ii) Multilayer Perceptron										4	2	2																																	
(b)	Compute the Linear Discriminant projection for the following dimensional data set Samples for class w_1 : $X_1 = \{(4,2), (2,4), (3,6), (4,4)\}$ Samples for class w_2 : $X_2 = \{(9,10), (6,8), (9.5), (8,7), (10,8)\}$										6	4	2																																	

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Q.NO		Part –A (5 x 2 = 10)	Marks	BT	CO																																								
1		Briefly explain any two of the following (i) Supervised learning (ii) Unsupervised learning (iii) Reinforced learning	2	1	1																																								
2		Define with suitable example “version space representation algorithm”.	2	1	1																																								
3		Define Bayes Rule	2	1	1																																								
4		Give decision tree to represent the following Boolean function (i) $A \wedge B$ (ii) $A \vee [B \wedge C]$	2	1	1																																								
5		Describe the FIND – S Algorithm	2	1	1																																								
Part –B (2 x 10 = 20)																																													
6 (a)		Summarize the choices in designing a Learning Example with Suitable Example.	8	6	2																																								
(b)		Define the inductive learning Hypothesis	2	1	1																																								
7 (a)		Consider the Enjoysport Learning task and assume the Hypothesis task H. Let General Hypothesis be $\langle ?, ?, ?, ?, ?, ? \rangle$ and Specific Hypothesis be defined as $\langle \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset \rangle$ <table border="1"><thead><tr><th>Example</th><th>Sky</th><th>Airtemp</th><th>Humidity</th><th>Wind</th><th>Water</th><th>Forecast</th><th>Enjoysport</th></tr></thead><tbody><tr><td>1</td><td>Sunny</td><td>Warm</td><td>Normal</td><td>Strong</td><td>Warm</td><td>Same</td><td>Yes</td></tr><tr><td>2</td><td>Sunny</td><td>Warm</td><td>High</td><td>Strong</td><td>Warm</td><td>Same</td><td>Yes</td></tr><tr><td>3</td><td>Rainy</td><td>Cold</td><td>High</td><td>Strong</td><td>Warm</td><td>Change</td><td>No</td></tr><tr><td>4</td><td>Sunny</td><td>Warm</td><td>High</td><td>Strong</td><td>Cold</td><td>Change</td><td>Yes</td></tr></tbody></table> Trace the CANDIDATE_ELIMINATION algorithm for a hypothesis in H' given by $\langle ?, \text{Cold}, \text{High}, ?, ?, ? \rangle \vee \langle \text{Sunny}, ?, \text{High}, ?, ?, \text{Same} \rangle$	Example	Sky	Airtemp	Humidity	Wind	Water	Forecast	Enjoysport	1	Sunny	Warm	Normal	Strong	Warm	Same	Yes	2	Sunny	Warm	High	Strong	Warm	Same	Yes	3	Rainy	Cold	High	Strong	Warm	Change	No	4	Sunny	Warm	High	Strong	Cold	Change	Yes	5	3	2
Example	Sky	Airtemp	Humidity	Wind	Water	Forecast	Enjoysport																																						
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(b)		State and Prove Maximum Likelihood Estimation Algorithm.	5	3	2																																								
(a)		State and Prove Maximum a posteriori probability Algorithm	5	3	2																																								
(b)		Consider the following set of training examples <table border="1"><thead><tr><th>Instance</th><th>Classification</th><th>A1</th><th>A2</th></tr></thead><tbody><tr><td>1</td><td>+</td><td>T</td><td>T</td></tr><tr><td>2</td><td>+</td><td>T</td><td>T</td></tr><tr><td>3</td><td>-</td><td>T</td><td>F</td></tr><tr><td>4</td><td>+</td><td>F</td><td>F</td></tr><tr><td>5</td><td>-</td><td>F</td><td>T</td></tr><tr><td>6</td><td>-</td><td>F</td><td>T</td></tr></tbody></table> What is the entropy of this collection w.r.t target function classification What is the information gain of A2 relative to this training example.	Instance	Classification	A1	A2	1	+	T	T	2	+	T	T	3	-	T	F	4	+	F	F	5	-	F	T	6	-	F	T	5	3	2												
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